

SKILL – 12:

PART 1 — Single Pipe:

Code:

```
GNU nano 7.2 skill12_pipe.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    int pipefd[2];
    pid_t pid;
    char buffer[100];

    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    pid = fork();

    if (pid == -1) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        // Child process - reads from pipe
        close(pipefd[1]);
        Read 58 lines
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$ gcc skill12_pipe.c -o skill12_pipe
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$ ./skill12_pipe
Child received: Hello from Parent through Pipe
Parent: Child process completed.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$
```

PART 2 — Multiple Pipes / Long Pipeline:

Code:

```
GNU nano 7.2 skill12_multipipe.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    int pipe1[2];
    int pipe2[2];

    pid_t p1, p2, p3;

    if (pipe(pipe1) == -1) {
        perror("pipe1");
        return 1;
    }

    if (pipe(pipe2) == -1) {
        perror("pipe2");
        return 1;
    }

    // First process
    p1 = fork();

    if (p1 == -1) {
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$ nano skill12_multipipe.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$ gcc skill12_multipipe.c -o skill12_multipipe
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$ ./skill12_multipipe
Process 3 received: Data from Process 1 -> Process 2
Parent: All processes completed successfully.
Parent: Pipes closed and resources cleaned up.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-12$
```